

(such as KMail) and your mail server. In POP, all the mails are downloaded to the user's PC.

4. Internet Mail Application Protocol (IMAP): This, for all practical purposes, does the same thing as POP, but there is one major technical difference. In IMAP, the mail is not downloaded to the user's PC, but is kept on the server itself. In essence, the mail client (such as KMail or Outlook) acts as an online browser for the mails on the server.
5. Web mail application: A Web mail application enables access to our mails using a Web interface. What we use at mail.google.com when we want to check our Gmail accounts is a Web mail interface to Google's POP, IMAP and SMTP servers.
6. Maildir: This is a file on the server that stores all the mail folders of an account.

So you see, we need to install two pieces of server software (an MTA and a POP/IMAP server) to actually get a working messaging server. If we install only an MTA, we will be able to send and receive mails, but not read them. And with only a POP-IMAP daemon, we will only be able to read mails, not send and receive them (without receiving capabilities, from where will we get mails to read?!

Getting started

Our software set is pretty mundane. We need an MTA and a POP-IMAP daemon. CentOS comes with Sendmail (the old and faithful MTA) installed by default, something we will do away with. The best choice for a MTA is QMail, but it's a pain to maintain. We will settle for Postfix. As for POP-IMAP, we will use the best—Dovecot.

We need some sort of an authentication mechanism, which will be provided by Cyrus-SASL. A final touch will be SpamAssassin and ClamAV, to deal with mail malice.

Installing the stuff

Open a terminal, and type the following two commands as the root user:

```
yum remove sendmail
yum install cyrus-sasl cyrus-sasl-devel cyrus-sasl-gssapi \
cyrus-sasl-md5 cyrus-sasl-plain postfix dovecot
```

This should pull in all the stuff we want, plus the dependencies.

The following section is completely based on a very helpful article at [HowToForge](#)—a link is presented at the *References* section.

Configuring the mail server

First of all, open up a terminal window. CentOS has a very annoying problem with its PATH variable; it doesn't include any of the *sbin* directories. So type in:

```
export PATH=$PATH:/sbin:/usr/sbin
```

Now to set up SASL-TLS authentication:

```
postconf -e 'smtpd_sasl_local_domain ='
postconf -e 'smtpd_sasl_auth_enable = yes'
postconf -e 'smtpd_sasl_security_options = noanonymous'
postconf -e 'broken_sasl_auth_clients = yes'
postconf -e 'smtpd_sasl_authenticated_header = yes'
postconf -e 'smtpd_recipient_restrictions = permit_sasl_authenticated,permit_mynetworks,reject_unauth_destination'
postconf -e 'inet_interfaces = all'
postconf -e 'mynetworks = 127.0.0.0/8'
```

We must edit `/usr/lib/sasl2/smtpd.conf` so that Postfix allows both 'PLAIN' and 'LOGIN' log-ins. The file should look like this:

```
pwcheck_method: saslauthd
mech_list: plain login
```

Now we need to create some self-signed certificates for encryption:

```
mkdir /etc/postfix/ssl
cd /etc/postfix/ssl/
openssl genrsa -des3 -rand /etc/hosts -out smtpd.key 1024
```

Remember this passphrase you were asked for. You'll need it extensively for the next steps.

```
openssl req -new -key smtpd.key -out smtpd.csr
openssl x509 -req -days 3650 -in smtpd.csr -signkey smtpd.key -out
smtpd.crt
openssl rsa -in smtpd.key -out smtpd.key.unencrypted
mv -f smtpd.key.unencrypted smtpd.key
openssl req -new -x509 -extensions v3_ca -keyout cakey.pem -out cacert.
pem -days 3650
```

Next we configure Postfix with these certificates:

```
postconf -e 'smtpd_tls_auth_only = no'
postconf -e 'smtp_use_tls = yes'
postconf -e 'smtpd_use_tls = yes'
postconf -e 'smtp_tls_note_starttls_offer = yes'
postconf -e 'smtpd_tls_key_file = /etc/postfix/ssl/smtpd.key'
postconf -e 'smtpd_tls_cert_file = /etc/postfix/ssl/smtpd.crt'
postconf -e 'smtpd_tls_cafile = /etc/postfix/ssl/cacert.pem'
postconf -e 'smtpd_tls_loglevel = 1'
postconf -e 'smtpd_tls_received_header = yes'
postconf -e 'smtpd_tls_session_cache_timeout = 3600s'
postconf -e 'tls_random_source = dev:/dev/urandom'
```

Then we set the hostname in our Postfix installation (make sure you replace mail.linux.bogus with your own hostname):

```
postconf -e 'myhostname = mail.linux.bogus'
```